The Framework in Brief

$$D = M/R \qquad \text{(inertial density)}$$
$$D_{crit} = c^2/2G = 6.732\,95\text{e}26\,\mathrm{kg\,m^{-1}} \qquad \text{(threshold)}$$
$$k = D/D_{crit} = r_s/R = v^2/c^2 \qquad \text{(scale factor)}$$
$$GTD = \sqrt{1-k} \qquad \text{(gravitational TD)}$$
$$DTD = \sqrt{k} \qquad \text{(DeGerlia TD)}$$
$$DTD^2 + GTD^2 = 1 \qquad \text{(unit relation)}$$

	Case 1 $\sim r_s$	Case 4 $\rho = \rho$	Case 7 classical
$\sqrt{D_1/D_2}$ (DeGerlia)	4.85439e2	1.00000e1	2.00000e0
$\sqrt{\frac{M_1 R_2}{M_2 R_1}}$ (Schwarzschild)	4.85439e2	1.00000e1	2.00000e0
GTD ratio (via r_s)	1-9.96838e-1	1-7.35193e-6	1-2.22785e-27
GTD ratio (via D_{crit})	1-9.96948e-1	1-7.35193e-6	1-2.22785e-27

Excerpt — full comparison across seven regimes in Table 2.